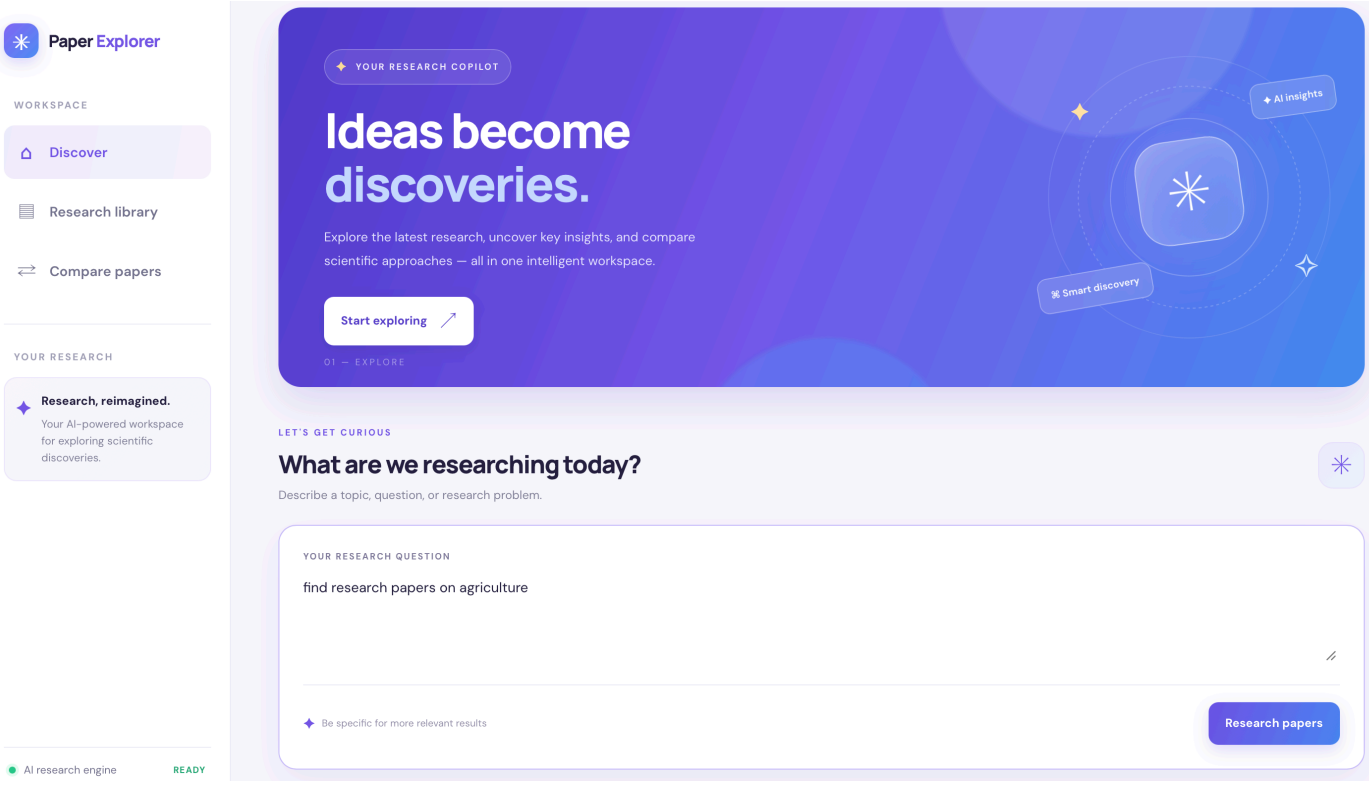
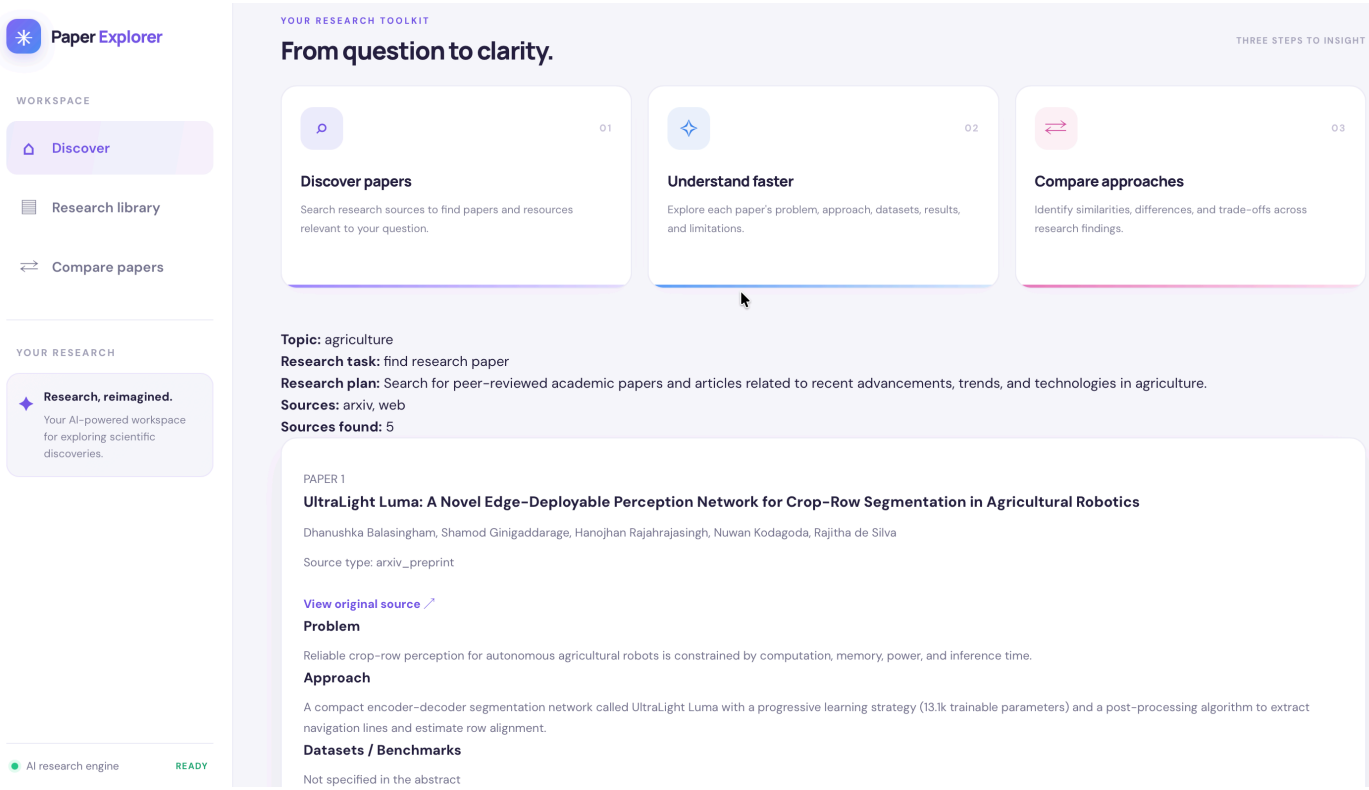

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PAPER 1
UltraLight Luma: A Novel Edge-Deployable Perception Network for Crop-Row Segmentation in Agricultural Robotics
Dhanushka Balasingham, Shamod Ginigaddarage, Hanojhan Rajahrajasingh, Nuwan Kodagoda, Rajitha de Silva
Source type: arxiv_preprint
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Problem
Reliable crop-row perception for autonomous agricultural robots is constrained by computation, memory, power, and inference time.
Approach
A compact encoder-decoder segmentation network called UltraLight Luma with a progressive learning strategy (13.1k trainable parameters) and a post-processing algorithm to extract navigation lines and estimate row alignment.
Datasets / Benchmarks
Not specified in the abstract

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Results

Improved parameter efficiency over U-Net, YOLOv8, and YOLOv26 while maintaining reliable crop-row detection performance, requiring only 25.23 mJ per inference.

Limitations

Not specified in the abstract

PAPER 2

Generative and deterministic deep learning models comparison for fine-scale precipitation retrievals from infrared brightness temperature

Matthieu Meignin, Cécile Mallet, Nicolas Viltard

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Problem

Accurate high-resolution precipitation estimation from infrared brightness temperatures is challenging because measurements characterize cloud-top properties rather than surface precipitation processes, limiting their correlation with rainfall intensity and leading to underestimation of extreme events.

Approach

A systematic inter-comparison of state-of-the-art deep learning models for high-resolution precipitation retrieval, including deterministic U-Nets, transformer-based architectures, conditional GANs, and diffusion models, along with tailored preprocessing and sampling strategies.

Datasets / Benchmarks

A curated dataset spanning 2008 to 2023 combining Météo-France radar mosaics as reference with multi-channel infrared observations from Meteosat Second Generation over metropolitan France.

Results

Deterministic models provide robust mean estimates and pixel-wise accuracy but underestimate extreme precipitation, whereas generative models better capture the full precipitation distribution, extreme events, and realistic spatial structures at the cost of reduced pixel-wise fidelity.

Limitations

Not specified in the abstract

PAPER 3

IVG-UAV: An Intelligent Voice-Guided UAV System for Autonomous Ripe Fruit Harvesting with Vision-Based Classification and Adaptive Path Planning

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Dinh Trung Duong

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Problem

Agricultural sectors in tropical regions depend heavily on manual labor for fruit harvesting, leading to significant labor costs and logistic complexities on large-scale farms.

Approach

Development and simulation of a voice-controlled Unmanned Aerial Vehicle (UAV) system integrating Whisper and LLM for speech recognition, computer vision-based ripeness classification, and adaptive path planning within a unified framework.

Datasets / Benchmarks

Not specified in the abstract

Results

The system was successfully modeled and validated in a Gazebo simulation environment, enabling performance evaluation under controlled agricultural scenarios.

Limitations

Not specified in the abstract

PAPER 4

Anaximander: Interactively Running Geospatial Deep Learning Models on Any Compute Backend

Satej S. Soman, Akram Zaytar, Girmaw A. Tadesse, Gilles Q. Hacheme, Muhammad S. Danish, Inbal Becker-Reshef, Rahul Dodhia, Juan Lavista Ferres

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Problem

Applying deep learning models to satellite imagery from within geographic information systems remains high-friction due to incompatible model formats, different compute environments, and the need for custom deployment, tiling, and georeferencing code.

Approach

An open-source system called Anaximander that unifies model source and compute location choice behind an interactive interface. It consists of an inference server backend that loads models from multiple sources, manages sessions, caches models, and streams results, paired with a QGIS plugin for tiling, reassembly, georeferencing, and status visualization.

Datasets / Benchmarks

Not specified in the abstract

Results

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Demonstrated a code-free side-by-side comparison of three heterogeneous models (gpt-image-1 via cloud API, SAM3 on a remote GPU, and DelineateAnything on a local CPU) on an agricultural field delineation task.

Limitations

Not specified in the abstract

PAPER 5

A Swarm-Coordinated Multi-Robot System for Early Stress Detection in Agricultural Rows Using Multimodal Leaf Sensing

Rishi Gupta, Astha Goyal, Vinay Vishwakarma

Source type: arxiv_preprint

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Problem

Early stress detection in crops is necessary to improve efficiency and reduce waste, but most modern techniques like hyperspectral imaging and AI-based systems are too costly and complex for medium and small-scale farmers.

Approach

CropSentry, a low-cost, ground-based multi-robot system using multimodal leaf sensing to continuously monitor crop health by tracking stress levels. The system comprises two autonomous bots that detect leaf color and environmental data row by row, spatially map observations, and send them to a master bot which generates a real-time web-based dashboard.

Datasets / Benchmarks

63 observations collected during experiments and 10 slave observations for wireless communication testing.

Results

Overall crop health classification accuracy of 84.12%, with 82.60% for healthy plants, 88% for nutrient-deficient plants, and 80% for diseased plants, along with a 100% wireless communication success rate across 10 slave observations.

Limitations

Not specified in the abstract

Overall comparison

The five papers cover diverse applications of machine learning, computer vision, and robotics, with a heavy emphasis on agricultural technology (Papers 1, 3, and 5), alongside geospatial deep learning infrastructure (Paper 4) and fundamental precipitation retrieval modeling (Paper 2). While Paper 1 focuses on extreme parameter efficiency for edge computing on agricultural

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communication success rate across 10 slave observations.

Limitations

Not specified in the abstract

Overall comparison

The five papers cover diverse applications of machine learning, computer vision, and robotics, with a heavy emphasis on agricultural technology (Papers 1, 3, and 5), alongside geospatial deep learning infrastructure (Paper 4) and fundamental precipitation retrieval modeling (Paper 2). While Paper 1 focuses on extreme parameter efficiency for edge computing on agricultural robots, Paper 5 addresses low-cost multi-robot swarm sensing for crop health, and Paper 3 integrates voice control and UAVs for harvesting. Paper 2 compares deterministic versus generative deep learning models for meteorology, and Paper 4 provides an open-source software integration layer (Anaximander) to run geospatial deep learning models seamlessly inside GIS software.

Similarities

- Multiple papers focus heavily on agricultural applications, specifically robotics, crop monitoring, and harvesting automation.
- Most papers incorporate deep learning or computer vision models to solve complex real-world data processing challenges.
- The abstracts consistently detail a structured approach comprising a problem statement, a newly proposed system or model, and validation metrics or results.

Differences

- Paper 1 targets ultra-lightweight edge deployment on agricultural robots, whereas Paper 2 compares heavy deterministic and generative deep learning models for meteorological analysis.
- Paper 3 develops an interactive voice-controlled UAV harvesting simulation, while Paper 5 relies on a physical ground-based multi-robot swarm using multimodal leaf sensors.
- Paper 4 focuses on infrastructure and interoperability by building a bridge between deep learning models and geographic information systems via QGIS, unlike the domain-specific algorithmic solutions in the other papers.

Suggested strongest approach

Paper 1 presents the strongest and most resource-efficient approach within its specific application domain, achieving high parameter efficiency (13.1k trainable parameters) and low energy consumption (25.23 mJ per inference) crucial for edge-deployable agricultural robotics, though the best approach ultimately depends on the distinct goals of each paper's respective domain.

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